

PIETARSAARI KALLAN, Finland

WMO: 029200

Lat: 63.751N

Lon: 22.523E

Elev: 19

StdP: 101.09

Time Zone: 2.00 (EUE)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-20.6	-17.0	-23.2	0.5	-20.5	-19.1	0.7	-16.0	19.0	-0.8	17.5	-0.1	4.1	110	0.879

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	3.8	22.5	18.1	20.7	17.1	19.3	16.3	19.2	21.8	17.9	20.2	16.7	18.9	5.2	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.1	20.9	16.8	12.0	19.4	15.6	11.1	18.2	54.6	21.9	50.5	20.2	47.0	19.0	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.7	14.9	13.4	DB	-20.5	24.7	4.6	2.5	-23.9	26.4	-26.6	27.9	-29.1	29.3	-32.5	31.0	(4)
(5)			WB	-20.9	19.8	4.8	1.9	-24.4	21.2	-27.2	22.3	-29.9	23.3	-33.5	24.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.8	-4.8	-5.6	-3.1	1.9	7.2	11.4	15.8	15.4	11.5	6.1	2.1	-1.1		
	DBStd	8.27	5.53	6.11	4.38	2.63	3.42	2.96	3.12	2.61	2.53	3.03	3.45	4.76		
	HDD10.0	2381	458	436	407	243	102	16	1	1	14	124	237	343		
	HDD18.3	4959	716	669	665	493	346	208	91	97	206	380	487	602		
	CDD10.0	484	0	0	0	0	14	59	181	168	58	3	0	0		
	CDD18.3	20	0	0	0	0	0	1	13	6	0	0	0	0		
	CDH23.3	29	0	0	0	0	0	1	17	12	0	0	0	0		
	CDH26.7	3	0	0	0	0	0	0	2	1	0	0	0	0		
(14) Wind	WSAvg	6.9	7.4	6.9	6.9	6.4	6.4	6.1	5.7	6.0	7.1	8.0	8.1	8.1		
(15) Precipitation	PrecAvg	528	39	27	29	26	37	50	67	57	52	56	49	43		
	PrecMax	674	69	50	49	66	70	87	121	132	110	95	113	83		
	PrecMin	386	10	3	5	6	5	13	5	20	20	15	5	17		
	PrecStd	78	16	15	12	15	19	21	31	32	28	22	24	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.0	3.7	5.2	12.5	20.4	21.7	25.1	24.8	18.3	13.1	9.2	6.0		
		MCWB	3.4	2.3	2.4	7.1	13.0	15.5	19.5	18.6	15.2	11.4	8.0	4.9		
	2%	DB	2.9	2.7	3.7	9.0	17.1	18.9	23.1	21.7	16.7	11.9	8.2	5.2		
		MCWB	2.0	1.6	1.9	6.0	12.1	14.7	19.1	17.0	14.2	10.5	7.3	4.2		
	5%	DB	2.1	1.8	2.9	6.3	14.1	17.3	21.5	20.0	15.6	10.9	7.3	4.7		
		MCWB	1.5	1.1	1.6	4.0	10.3	13.8	18.2	16.7	13.6	9.7	6.3	3.7		
	10%	DB	1.4	1.1	2.2	5.0	11.8	15.6	20.1	18.8	14.8	10.1	6.4	3.8		
		MCWB	0.8	0.4	1.0	3.3	8.8	12.6	17.2	16.0	13.1	8.9	5.4	2.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.7	2.7	3.2	8.8	14.6	17.1	20.9	19.6	15.9	11.8	8.2	5.5		
		MCDB	3.9	3.4	4.6	10.2	18.7	20.3	23.1	23.2	17.3	12.6	8.8	5.8		
	2%	WB	2.3	1.8	2.4	5.8	12.5	15.5	19.9	18.1	14.9	10.6	7.2	4.6		
		MCDB	2.6	2.5	3.4	7.7	16.8	18.1	22.7	20.7	16.2	11.4	7.9	5.0		
	5%	WB	1.6	1.3	1.9	4.2	10.7	14.2	18.9	17.1	14.0	9.7	6.2	3.9		
		MCDB	2.0	1.8	2.8	5.6	14.1	16.8	21.3	19.3	15.1	10.5	6.7	4.6		
	10%	WB	0.8	0.7	1.2	3.3	9.2	12.8	17.7	16.4	13.3	8.8	5.4	3.1		
		MCDB	1.3	1.1	2.1	4.6	11.7	15.1	19.9	18.3	14.6	9.7	6.0	3.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.2	4.3	4.5	3.9	4.8	4.1	3.8	3.5	3.3	3.0	2.9	3.3		
		MCDBR	3.2	3.2	3.7	6.0	9.0	6.7	5.5	5.2	4.0	3.4	2.9	2.9		
		MCWBR	3.0	2.9	2.9	3.7	5.1	3.9	3.2	3.0	2.7	3.3	3.0	2.8		
		MCWBR	3.0	3.0	3.6	5.1	8.6	6.2	5.0	4.1	3.5	3.5	3.0	2.9		
(40) Clear-Sky Solar Irradiance	taub	taub	0.228	0.287	0.312	0.351	0.350	0.351	0.371	0.358	0.333	0.314	0.364	0.351		
		taud	2.021	2.149	2.208	2.267	2.441	2.483	2.452	2.481	2.528	2.373	2.616	2.571		
		Ebn at Noon	358	623	772	823	864	870	837	816	766	603	330	178		
		Edh at Noon	28	62	89	105	97	95	96	85	67	51	26	16		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	0.14	0.70	2.21	3.80	5.13	5.67	5.18	3.95	2.26	0.84	0.19	0.06		
		RadStd	0.01	0.14	0.23	0.34	0.52	0.49	0.48	0.38	0.23	0.15	0.03	0.01		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page